

COMP CO835 Object-Oriented Systems

Class Exercise 7 (Lab 7)

You are tasked with designing a Hospital Management System (HMS) for a multi-specialty hospital. The HMS should facilitate the management of patient appointments, medical records, and billing processes. The system should be user-friendly, efficient, and ensure data security and integrity.

The system will include the following features or requirements:

1. **Patient Management:**
 - Register new patients.
 - Schedule patient appointments with doctors.
 - Maintain patient demographics and medical history.
2. **Appointment Management:**
 - Schedule appointments for patients with specific doctors.
 - Allow patients to reschedule or cancel appointments.
3. **Doctor Management:**
 - Register doctors with their specialization and availability.
 - Assign doctors to patient appointments.
4. **Billing Management:**
 - Generate bills for services and treatments provided to patients.
 - Accept payments from patients and provide receipts.

Task 1: Use Case Diagram

- Identify and list the actors and use cases in the system. (e.g., Schedule Appointment, Add Patient, View Patient Records).
- Create a use case diagram that visually represents the relationships between actors and use cases.
- Include generalization and include/extend relationships where necessary.

Task 2: Activity Diagram

- Select one use case from the use case diagram.
- Create an activity diagram to model the step-by-step process within the selected use case.
- Include decision points, forks, and joins as needed to represent the flow of activities.
- Ensure that the activity diagram is clear and follows a logical sequence.